Maths IA-1 — THE ONLY SHEET (one night, one source, final)

> **CLOSE EVERYTHING ELSE.** This file is your entire night and your entire exam prep: schedule · 27 quick traps · 8 hard questions · 4 applications · playbook · formula card. Every question now has: **CONCEPT (kid-language)** (the idea explained like you're 10 — no prior knowledge needed) · **TRIGGER** (how the paper signals this type) · **STEPS** (full solution) · **REMEMBER** (the line that scores the mark).

NIGHT SCHEDULE — follow the clock, no decisions needed

Block	Time	Do
I	now – +2.5h	TIER 1: 27 quick traps (read TRIGGER → cover STEPS → solve → check)
—	+2.5h – +2.75h	break: walk, water, food
II	+2.75h – +6h	TIER 2 (8 hard) + TIER 3 (4 applications), full steps
III	+6h – +6.5h	re-solve EVERY wrong one with the answer covered
IV	+6.5h – +7.5h	memorise FORMULA CARD + PLAYBOOK
V	+7.5h – +8.5h	timed re-run Tier 1, 25 min, zero errors
VI	+8.5h – +9h	timed re-run 5 best Tier 2, 30 min
VII	final 30 min	formula card glance, get ready, leave

TIER 1 — 27 quick-mark traps (the safe marks in ANY format)

1. If $a < b$, which need NOT be true? (a) $a-3 < b-3$ (b) $-4a > -4b$ (c) $a^2 < b^2$ (d) $3a < 3b$

- **CONCEPT (kid-language):** $a < b$ just means a is the smaller number. Add the same to both sides → order stays the same. Multiply both by a NEGATIVE → the order flips (like a mirror image). Squaring only keeps the order if both numbers are positive — that's the trap.
- **TRIGGER:** wording "need NOT"/"always" → test with $a = -5$, $b = 1$
- **STEPS:** (a) $-8 < -2$ ✓ · (b) $20 > -4$ ✓ (flip works) · (c) $25 < 1$ ✗ FALSE → answer **(c)** · (d) $-15 < 3$ ✓
- **REMEMBER:** × negative flips, × positive keeps, **squaring breaks order when signs differ**

2. $\log_2(x-1) = 3 \rightarrow x = ?$

- **CONCEPT (kid-language):** A log is just the question: '2 multiplied by itself how many times gives this number?' $\log_2(8) = 3$ because $2\times2\times2 = 8$. So your equation is saying $2\times2\times2 = x-1$, meaning $x-1 = 8$, so $x = 9$.
- **TRIGGER:** log of an expression = number → drop the log first
- **STEPS:** $x - 1 = 2^3 = 8 \rightarrow x =$ **9**
- **REMEMBER:** undo log BEFORE touching the +1; **8 is the bait answer**

3. $x^2 + (k-3)x + k = 0$ has equal roots → sum of possible k = ? product = ?

- **CONCEPT (kid-language):** Every quadratic $x^2+bx+c = 0$ has one magic number: $D = b^2-4ac$. $D > 0 \rightarrow$ two answers. $D = 0 \rightarrow$ exactly one (they're 'equal'). $D < 0 \rightarrow$ no answers. The words 'equal roots' or 'touches the axis' mean ONLY ONE job: write $D = 0$ and solve it.
- **TRIGGER:** "equal roots" / "touches the axis once"
- **STEPS:** $D = (k-3)^2 - 4k = 0 \rightarrow k^2 - 6k + 9 - 4k = k^2 - 10k + 9 = 0 \rightarrow k = 1, 9 \rightarrow$ sum = **10**, product = **9**
- **REMEMBER:** $D = 0$ engine; **underline sum vs product BEFORE solving**

4. Solution of $(x - 3)/(x + 1) < 0 \rightarrow$

- **CONCEPT (kid-language):** A fraction is negative only when the TOP and BOTTOM have opposite signs (one +, one –). Find where the top becomes 0 ($x = 3$) and where the bottom becomes 0 ($x = -1$) — those two points split the number line — then test a number in between.
- **TRIGGER:** rational $< 0 \rightarrow$ numerator and denominator opposite signs $\rightarrow$ between the roots
- **STEPS:** roots -1 and $3 \rightarrow$ test $x = 0$: $(-3)/(1) = -3 < 0$ ✓ inside $\rightarrow$ **$-1 < x < 3$**, $x \neq -1$
- **REMEMBER:** denominator point NEVER included; no flip ever happens here

5. Digits of 2^{100} ($\log 2 = 0.3010$) $\rightarrow$

- **CONCEPT (kid-language):** $\log_{10}(N)$ = 'ten times itself how many times reaches N?'. If that answer is 30.10, you have 30 full jumps plus a tiny bit more $\rightarrow$ the number has 31 digits. Rule: digits = the whole-number part + 1.
- **TRIGGER:** "how many digits"
- **STEPS:** $\log(2^{100}) = 30.10 \rightarrow \lfloor 30.10 \rfloor + 1 =$ **31**
- **REMEMBER:** digits = $\lfloor \log N \rfloor + 1$ — the **+1** is the whole question

6. Last two digits of 7^{200} $\rightarrow$

- **CONCEPT (kid-language):** 'Last two digits' means 'remainder after dividing by 100'. Don't compute 7^{200} — notice $7\times7\times7\times7 = 2401$ ends in 01, and $01 \times 01 \times 01$... always stays 01. So the answer is 01.
- **TRIGGER:** "last two digits" $\rightarrow$ mod 100
- **STEPS:** $7^4 = 2401 \equiv 01 \pmod{100} \rightarrow 7^{200} = (7^4)^{50} \equiv 1^{50} =$ **01**
- **REMEMBER:** memorise hero $7^4 \equiv 01$ — one line, full marks

7. gcd(a,b) = 12, lcm(a,b) = 120 $\rightarrow$ ordered pairs (a,b) = ?

- **CONCEPT (kid-language):** GCD = the biggest block that evenly divides BOTH numbers. LCM = the smallest number both fit into. Golden rule to memorise: number × number = GCD × LCM, always true, like a law of gravity.
- **TRIGGER:** gcd AND lcm both given $\rightarrow$ pull the gcd out
- **STEPS:** $a = 12x$, $b = 12y$ with $xy = 10$ and $\gcd(x,y) = 1 \rightarrow (1,10)(2,5)(5,2)(10,1) \rightarrow$ **4**
- **REMEMBER:** divide both by the gcd first, then coprime-factor the rest

8. "All multiples of 4 are even. 12 is even. So 12 is a multiple of 4." Why invalid?

- **CONCEPT (kid-language):** 'Every dog is an animal. This is an animal. So it's a dog' — silly, right? It could be a cat. That's the flaw: being even doesn't make 12 a multiple of 4, because 6 is even and NOT a multiple of 4. One counterexample kills the logic.
- **TRIGGER:** "reasoning incorrect/flawed" $\rightarrow$ find counterexample
- **STEPS:** 6 is even but not a multiple of 4 $\rightarrow$ the conclusion doesn't follow $\rightarrow$ **(b)**
- **REMEMBER:** going backwards from a property needs proof, not vibes — cite the counterexample

9. $|x| = -x$ exactly when $\rightarrow$

- **CONCEPT (kid-language):** $|x|$ = how far x sits from 0 on the number line — it's never negative. $|x| = -x$ works only when x is 0 or below (then $-x$ is a positive number). Trap: don't forget 0 — because $|0| = -0$ also works.
- **TRIGGER:** absolute value equals its negative
- **STEPS:** $x = 0$: $0 = 0$ ✓; $x < 0$: $|x| = -x$ ✓; $x > 0$: $x = -x$ ✗ $\rightarrow$ **$x \leq 0$**
- **REMEMBER: 0 is included** — trap answer is $x < 0$

10. $x + 1/x = 4 \rightarrow x^3 + 1/x^3 = ?$

- **CONCEPT (kid-language):** There's a family of ready-made shortcuts for x + 1/x. The cube one: $(x+1/x)^3 = x^3+1/x^3 + 3(x+1/x)$. If x+1/x = 4: 64 = answer + 3×4 → answer = 64 − 12 = 52. Just memorise the recipe and pour in the number.
- **TRIGGER:** x + 1/x given, higher power asked
- **STEPS:** $(x+1/x)^3 = x^3 + 1/x^3 + 3(x+1/x) \rightarrow 64 = ans + 12 \rightarrow$ **52**
- **REMEMBER:** cube formula subtracts **3×(the given)**, not 3

11. $\log_5 125 + \log_2 32 - \log_{10} 10000 = ?$

- **CONCEPT (kid-language):** Each of these logs is a 'what power?' question: 5^? = 125 → 3; 2^? = 32 → 5; 10^? = 10000 → 4. Turn each into its power first, THEN do the arithmetic: 3 + 5 − 4 = 4.
- **TRIGGER:** logs of exact powers → convert to exponents
- **STEPS:** $5^3=125 \rightarrow 3 \cdot 2^5=32 \rightarrow 5 \cdot 10^4=10000 \rightarrow 4 \rightarrow 3 + 5 - 4 =$ **4**
- **REMEMBER:** rewrite each as "to what power" — arithmetic then trivial

12. Remainder of $10^{10} \div 7 = ?$

- **CONCEPT (kid-language):** Remainders repeat in loops. For 3 dividing 7, the loop length is 6 (because $3^6 = 729$ leaves remainder 1). So cut the big exponent: 10 ÷ 6 leaves remainder 4 → same answer as 3^4. 81 ÷ 7 leaves 4.
- **TRIGGER:** big power remainder → reduce base, find cycle
- **STEPS:** $10 \equiv 3 \pmod 7$; $3^6 = 729 \equiv 1$; $10 = 6\times1 + 4 \rightarrow 3^4 = 81 = 7\times11 + 4 \rightarrow$ **4**
- **REMEMBER:** reduce base first, then exponent mod cycle length (6 here)

13. Divisors of 360 = ?

- **CONCEPT (kid-language):** For 360 = 2×2×2 × 3×3 × 5: the 2s can be picked 0, 1, 2 or 3 times → 4 choices; the 3s → 3 choices; the 5 → 2 choices. Multiply the choices: 4 × 3 × 2 = 24 ways → 24 divisors. Never count them one by one.
- **TRIGGER:** "number of divisors" → prime-factorise
- **STEPS:** $360 = 2^3 \cdot 3^2 \cdot 5 \rightarrow (3+1)(2+1)(1+1) =$ **24**
- **REMEMBER:** τ = product of (exponent + 1) — always +1

14. $(x - 2)$ divides $x^3 - kx^2 + 10x - 8$ exactly → k = ?

- **CONCEPT (kid-language):** '(x−2) divides exactly' means: when x = 2, the whole expression collapses to 0. So plug x = 2 in: 8 − 4k + 20 − 8 = 0 → 20 = 4k → k = 5. That's the entire trick.
- **TRIGGER:** "(x − a) divides/exactly" → factor theorem
- **STEPS:** f(2) = 8 − 4k + 20 − 8 = 0 → 20 − 4k = 0 → k = **5**
- **REMEMBER:** plug the ROOT (sign flips: x−2=0 → x=+2), set whole thing = 0

15. $4^{(x+1)} = 8^x \rightarrow x = ?$

- **CONCEPT (kid-language):** 4 and 8 are both built from 2s ($4 = 2^2$, $8 = 2^3$). Rewrite both sides with base 2: $2^{(2x+2)} = 2^{(3x)}$. If the bases are the same, the exponents MUST match → 2x+2 = 3x → x = 2.
- **TRIGGER:** power = power → make bases SAME
- **STEPS:** $2^{(2x+2)} = 2^{(3x)} \rightarrow 2x + 2 = 3x \rightarrow x =$ **2**
- **REMEMBER:** bases equal ⟹ exponents equal — never subtract bases

16. Product of roots of $3x^2 - 7x + 2 = 0 \rightarrow$

- **CONCEPT (kid-language):** You can know the roots' sum and product WITHOUT finding the roots: sum = −b/a, product = c/a (they come from expanding (x−α)(x−β)). For $3x^2-7x+2$: product = 2/3. Five seconds, done.
- **TRIGGER:** roots asked WITHOUT solving → Vieta's formulas
- **STEPS:** product = c/a = **2/3** (sum would be 7/3)
- **REMEMBER:** product = c/a, sum = −b/a — sign is the trap

17. MATCH: a) x+1/x=3 → x^2+1/x^2 · b) lcm(12,18) · c) 100 mod 7 · d) $\log_9 27$ · e) divisors of 60 — vs 7 / 36 / 2 / 3/2 / 12

- **CONCEPT (kid-language):** Matching = 5 tiny separate sums. Do the ones you recognise FIRST, lock their pairs, don't let one weird pair block the five easy ones. One pair at a time, independently.
- **TRIGGER:** matching → solve each pair INDEPENDENTLY, never chain
- **STEPS:** a) 9 – 2 = **7** · b) **36** · c) 100 = 14×7 + 2 → **2** · d) 9^{3/2}=27 → **3/2** · e) $2^2 \cdot 3 \cdot 5$ → 2·2·2 = **12**
- **REMEMBER:** pair-by-pair, lock the confident 3 first

18. Complete the square: $x^2 - 8x + 5$ →

- **CONCEPT (kid-language):** 'Make it a clean square': take half the middle number (–8 → –4), square it (16), write $(x-4)^2$, then subtract 16 back because you snuck it in: $(x-4)^2 - 16 + 5 = (x-4)^2 - 11$. Wobbly quadratic becomes tidy.
- **TRIGGER:** "write in the form $(x + p)^2 + q$"
- **STEPS:** half of –8 = –4 → $(x - 4)^2 - 16 + 5$ = **$(x - 4)^2 - 11$**
- **REMEMBER:** halve, square, SUBTRACT what you added

19. gcd(1071, 462) with Euclidean steps →

- **CONCEPT (kid-language):** Keep dividing the BIGGER by the SMALLER and keep only what's left over: 1071 ÷ 462 → left 147; then 462 ÷ 147 → left 21; then 147 ÷ 21 → nothing left. Stop. The last leftover — 21 — is your gcd. That's the whole Euclidean algorithm.
- **TRIGGER:** "use Euclidean algorithm" / big numbers
- **STEPS:** 1071 = 2×462 + 147 → 462 = 3×147 + 21 → 147 = 7×21 + 0 → **21**
- **REMEMBER:** write every division line — the steps are the marks, answer alone ≈ half credit

20. Integers 1–100 divisible by 6 OR by 8 →

- **CONCEPT (kid-language):** Multiples of 6 up to 100: 16 of them. Multiples of 8: 12. But numbers like 24, 48 (multiples of BOTH) were counted in each pile — so subtract the 4 double-counted: 16 + 12 – 4 = 24. That subtraction is the entire idea.
- **TRIGGER:** "or" in counting → inclusion–exclusion
- **STEPS:** $\lfloor 100/6 \rfloor = 16$, $\lfloor 100/8 \rfloor = 12$, $\lfloor 100/24 \rfloor = 4$ → 16 + 12 – 4 = **24**
- **REMEMBER:** $|A| + |B| - |A \cap B|$ — subtracting the double-count IS the question

21. Find k so $x^2 + 2x + k > 0$ for EVERY real x →

- **CONCEPT (kid-language):** 'Always bigger than 0' means the curve floats ABOVE the line and never touches it. Touching = D = 0 (fails), floating = D < 0 (wins). So 4 – 4k < 0 → k > 1. Note: k = 1 touches at one point → NOT > 0 → fails.
- **TRIGGER:** "for all real x" / "always positive"
- **STEPS:** need D < 0 and open-up: 4 – 4k < 0 → **k > 1**
- **REMEMBER:** strict — k = 1 fails (x = –1 gives exactly 0, not > 0)

22. Factorise fully: $x^4 - 16$ →

- **CONCEPT (kid-language):** A big square minus a small square splits: $A^2 - B^2 = (A-B)(A+B)$. So $x^4 - 16 = (x^2)^2 - 4^2 = (x^2-4)(x^2+4)$; then x^2-4 splits again = $(x-2)(x+2)$. The (x^2+4) piece can't split — that's your stop sign.
- **TRIGGER:** "fully" → apply difference of squares until it stops
- **STEPS:** $x^4 - 16 = (x^2)^2 - 4^2 = (x^2-4)(x^2+4)$ = **$(x-2)(x+2)(x^2+4)$**
- **REMEMBER:** $x^2 + 4$ never factors over reals — stop there

23. $2^3 \cdot 2^5 \div 2^4$ = →

- **CONCEPT (kid-language):** Same base? Only the POWERS move: multiplying adds powers, dividing subtracts. The base 2 never changes: $2^3 \cdot 2^5 \div 2^4 = 2^{(3+5-4)} = 2^4 = 16$. Never multiply the bases themselves.
- **TRIGGER:** same base, × ÷ → add/subtract exponents

- **STEPS:** $3 + 5 - 4 = 4 \rightarrow 2^4 =$ **16**
- **REMEMBER:** never multiply the bases — only the powers move

24. Express log 45 using log 3 and log 5 →

- **CONCEPT (kid-language):** Logs turn × into +: log(9 × 5) = log 9 + log 5. And log 9 = log 3^2 = 2 log 3. So log 45 = 2 log 3 + log 5. First FACTOR the number, then the logs fall apart.
- **TRIGGER:** "in terms of" → factorise the number
- **STEPS:** $45 = 9 \times 5 = 3^2 \cdot 5 \rightarrow 2 \log 3 + \log 5$
- **REMEMBER:** split the NUMBER, not the log

25. If $2^x = 7$ then x = →

- **CONCEPT (kid-language):** $2^x = 7$ and $x = \log_2 7$ are the exact same sentence — one spoken in power-language, one in log-language. The base of the power becomes the base of the log. Always. No exceptions.
- **TRIGGER:** exponential given, log form asked
- **STEPS:** $a^b = c \iff b = \log_a c \rightarrow x =$ **$\log_2 7$** (or log 7 / log 2)
- **REMEMBER:** base of log = base of power, always

26. gcd(a,b) = 8, lcm(a,b) = 96, a = 24 → b = ?

- **CONCEPT (kid-language):** If both numbers are built on the same block (8) and the shelf is 96: b = (8 × 96) ÷ 24 = 32. Rule: a × b = gcd × lcm — rearrange it like a formula and plug in.
- **TRIGGER:** three of the four gcd/lcm quantities → product rule
- **STEPS:** $a \cdot b = \text{gcd} \cdot \text{lcm} \rightarrow 24 \cdot b = 8 \times 96 = 768 \rightarrow b =$ **32**
- **REMEMBER:** gcd × lcm = a × b — the one law that binds them

27. Buses leave every 15 min and 20 min, both at 8:00 → next together at?

- **CONCEPT (kid-language):** 'Together' = the smallest number both intervals fit into = LCM. LCM(15, 20) = 60 minutes → 8:00 + 60 = 9:00. Whenever you see 'at the same time again', it's screaming LCM at you.
- **TRIGGER:** "next time together" → LCM of the intervals
- **STEPS:** lcm(15, 20) = 60 → 8:00 + 60 min = **9:00**
- **REMEMBER:** together = LCM, ever-decreasing gaps = HCF — never mix them

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TIER 2 — 8 hard questions (the heavy-marks engine; exam needs 2–3 perfect)

Full-mark rule: GIVEN → FORMULA → STEPS → ANSWER → one sentence of meaning.

H1. Last two digits of 31^{31}.

- **CONCEPT (kid-language):** 31 = 30 + 1. Expanding $(30+1)^{31}$ piece by piece: after the first two pieces, EVERY piece contains 30×30 = 900, whose last two digits are 00 — they disappear. Only 1 + 31×30 = 931 survives → last two digits 31.
- **TRIGGER:** last two digits of a NON-8-or-7 base → binomial split
- **STEPS:** $31 = 30 + 1 \rightarrow (30+1)^{31} = 1 + 31 \times 30 +$ terms with $30^2 = 900 \equiv 0 \pmod{100} \rightarrow 1 + 930 = 931 \rightarrow$ **31**
- **REMEMBER:** split into (tens + ones); everything with 30^2 dies mod 100

H2. Prove $n^4 - n^2$ divisible by 12 for every integer n.

- **CONCEPT (kid-language):** Proving divisible by 12? Split 12 into 3 × 4 and prove each part. Any three numbers in a row always include a multiple of 3 → part 1 done. For 4: if n is even, n^2 supplies both 2s; if n is odd, both neighbours are even → two 2s. Both parts proven → divisible by 12.
- **TRIGGER:** "prove divisible by" → factorise into consecutive products
- **STEPS:** $n^2(n-1)(n+1)$: $(n-1)n(n+1)$ gives a factor 3; n even → n^2 gives 4, n odd → two even neighbours give 4 → divisible by 3 AND 4, coprime → **12** ∎
- **REMEMBER:** split the divisor (12 = 3 × 4) and prove each part separately

H3. Greatest number dividing 552, 967, 1032 leaving the SAME remainder?

- **CONCEPT (kid-language):** If three numbers leave the SAME leftover when divided by N, subtracting pairs wipes the leftover away — so N must divide every difference. Answer = biggest number dividing all differences = 5. Then check: leftovers are 2, 2, 2 ✓ — the check IS a mark.
- **TRIGGER:** "same remainder" → gcd of DIFFERENCES
- **STEPS:** differences 415, 65, 480 → Euclid(415,65): 415 = 6×65+25 → 65 = 2×25+15 → 25 = 15+10 → 15 = 10+5 → **5**; check 552=110×5+2, 967=193×5+2, 1032=206×5+2 all leave 2 ✓
- **REMEMBER:** answer is gcd(differences) — **always verify remainders match** (that check is one mark)

H4. Same distance upstream in 4 h as downstream in 2 h; stream 3 km/h. Boat speed & distance?

- **CONCEPT (kid-language):** Same distance both ways → speed × time on both sides. Upstream speed = boat − current (b − 3), downstream = b + 3: 4(b−3) = 2(b+3) → 4b − 12 = 2b + 6 → b = 9. Distance = 4 × 6 = 24 km. Always verify with the other side: 2 × 12 = 24 ✓.
- **TRIGGER:** two trips, same distance → set the products equal
- **STEPS:** 4(b − 3) = 2(b + 3) → 4b − 12 = 2b + 6 → 2b = 18 → b = **9 km/h**; distance = 4 × 6 = **24 km**; check 2 × 12 = 24 ✓
- **REMEMBER:** with-stream = +, against = −; **write the check line**

H5. Solve log x + log(x + 3) = 1.

- **CONCEPT (kid-language):** Logs only accept POSITIVE numbers. First mash the logs together: log x + log(x+3) = log[x(x+3)] → quadratic → x = −5 or 2. Then THROW AWAY −5 (it would make log negative) → answer 2. Writing 'rejected because…' earns a mark.
- **TRIGGER:** logs added → combine → solve quadratic → DOMAIN CHECK
- **STEPS:** log[x(x+3)] = 1 → $x^2 + 3x = 10$ → $x^2 + 3x - 10 = 0$ → (x+5)(x−2) = 0 → x = −5 or 2; need x > 0 → **x = 2**
- **REMEMBER:** rejection line scores a mark — "x = −5 rejected, log undefined for x < 0"

H6. Solve 5x ≡ 3 (mod 8).

- **CONCEPT (kid-language):** Inverse = the thing that multiplies back to 1. In ÷8 world, 5 × 5 = 25 = 24 + 1 ≡ 1, so inverse of 5 is 5. Multiply both sides of 5x ≡ 3 by 5: x ≡ 15 ≡ 7 (mod 8).
- **TRIGGER:** linear congruence → need the inverse
- **STEPS:** 5×5 = 25 ≡ 1 (mod 8) → multiply both sides by 5: x ≡ 15 ≡ 7 → **x ≡ 7 (mod 8)**
- **REMEMBER:** find what multiplies to ≡ 1 — try small numbers: 1,3,5,7 (odd residues are self-inverse here)

H7. Four bells toll together at noon, then every 6, 8, 12, 15 s. Together how many times in 1 hour including start?

- **CONCEPT (kid-language):** All four together first happen after lcm(6, 8, 12, 15) = 120 seconds. In 3600 s: 3600 ÷ 120 = 30 times AFTER the start. Count the starting moment too → 31. Decide the +1 explicitly — that's the mark.
- **TRIGGER:** "together" + intervals → LCM + count including t = 0
- **STEPS:** lcm(6,8,12,15) = 120 s → 3600 ÷ 120 = 30 → +1 for the start = **31**
- **REMEMBER:** decide inclusion explicitly ("including the start") — +1 or not is the mark

H8. Sum of a number and its reciprocal is 10/3. Find it.

- **CONCEPT (kid-language):** Kill the fraction: multiply everything by 3x → $3x^2 + 3 = 10x$ → $3x^2 - 10x + 3 = 0$ → (3x−1)(x−3) = 0 → x = 3 or 1/3. Reciprocal answers come in PAIRS — give both, don't pick a favourite.

- **TRIGGER:** "sum of a number and its reciprocal" → clear fractions
- **STEPS:** $x + 1/x = 10/3 \rightarrow 3x^2 - 10x + 3 = 0 \rightarrow (3x-1)(x-3) = 0 \rightarrow x = 3$ or $1/3$ → **both**
- **REMEMBER:** word problems can have TWO answers — give both, don't pick

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TIER 3 — 4 application questions (central teams LOVE worded problems)

T1. Garden is 5 m longer than wide; area 36 m². Dimensions? Widened to 40 m² → width?

- **CONCEPT (kid-language):** '5 longer than wide' → width = w, length = w + 5. Area: $w(w+5) = 36$ → quadratic → $w = 4$ (reject -9 — a garden can't be -9 wide). Second part: same steps with 40 → $w \approx 4.3$ m. Words → one variable → quadratic → reject nonsense.
- **TRIGGER:** area word problem → quadratic
- **STEPS:** $w(w+5) = 36 \rightarrow w^2 + 5w - 36 = 0 \rightarrow (w+9)(w-4) = 0 \rightarrow w =$ **4 m, l = 9 m** (reject -9); then $w^2 + 5w - 40 = 0 \rightarrow w = (-5 + \sqrt{185})/2 \approx$ **4.3 m**
- **REMEMBER:** reject the negative length in a sentence, every time

T2. $IQ = MA/CA \times 100$. Age 15, $90 \le IQ \le 150$ → mental-age range?

- **CONCEPT (kid-language):** The formula IS your instructions: MA = IQ × age ÷ 100. Feed it the two ENDPOINTS: 90 × 15/100 = 13.5 and 150 × 15/100 = 22.5. Mental age sits between 13.5 and 22.5 years. Endpoints in, endpoints out.
- **TRIGGER:** chained inequality with a formula
- **STEPS:** $MA = IQ \times 15/100 \rightarrow 90 \rightarrow 13.5$, $150 \rightarrow 22.5 \rightarrow$ **$13.5 \le MA \le 22.5$ years**
- **REMEMBER:** substitute the ENDPOINTS only — inequalities map to endpoints

T3. $h(t) = -5t^2 + 20t + 25$: (i) zeros (ii) max height no calculus (iii) $h(6) = -35$ — "35 m underground"?

- **CONCEPT (kid-language):** Height formula: set h = 0 to find when it lands → t = 5 (reject $t = -1$: time can't be negative). Peak at $t = -b/2a = 2 \rightarrow 45$ m. And $h(6) = -35$ does NOT mean underground — the formula only works while flying, like a graph that stops when the rocket lands.
- **TRIGGER:** physics worded quadratic → always interpret in context
- **STEPS:** (i) $-5(t-5)(t+1) \rightarrow t =$ **5 s** (reject -1: time ≥ 0) · (ii) $t = -b/2a = 2 \rightarrow h(2) = 45 \rightarrow$ **45 m** · (iii) **NO** — formula valid only during flight; $t = 6$ is past landing, negative height is model extrapolation
- **REMEMBER:** interpretation questions need a full SENTENCE — the "why" is the mark

T4. $pH = -\log_{10}[H^+]$. A: 10^{-3}, B: $10^{-4.5}$. Which more acidic, by how much?

- **CONCEPT (kid-language):** The formula in the question IS the answer key: $pH = -\log(10^{-3}) = 3$, other = 4.5. Lower pH = more acidic → A wins by 1.5. Always say WHICH direction means 'more' — comparison needs a sentence, not just a number.
- **TRIGGER:** log in a real formula → compute both, then COMPARE
- **STEPS:** pH A = 3, pH B = 4.5 → lower pH = more acidic → **A by 1.5 units**
- **REMEMBER:** state direction ("lower pH is more acidic") — comparison needs reasoning, not just numbers

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PLAYBOOK — extracting 70% from an unpredictable, unknown-format, unknown-marks paper

Universal order:

1. **First 5 min:** read the WHOLE paper, note today's format, count what exists. No solving.
2. **Safe pass (~⅓ of time):** every question answerable in ≤ 90 s — in ANY order. This is your floor.
3. **Matching (if present):** pair independently, lock 3+ first.
4. **Heavy pass (~half):** pick 2–3 whose chapter you recognise → full steps. Then fight the rest.
5. **Final ~10%:** every blank gets SOMETHING; check signs, rejections, units.

70% checkpoint (mid-exam): safe pass locked + two heavy ones fully written = on track. If behind → heavy ones next, never sideways.

Twist protocol (confusing wording): read twice → **underline what's actually asked** (sum? product? "need NOT"?) → name the chapter → write givens + the one formula → solve. Still lost? Write formula + substitution anyway — partial credit is real credit.

Format-switch rules: MCQ with no stated negative marking → never blank, eliminate 2 first · "explain/justify" → full sentence naming the reason (domain, flip, sign, rejection, converse) · two candidate answers → plug into the ORIGINAL statement · freak format → it's chapters 1–6 underneath: identify → formula card.

3-minute rule: not cracked in 3 min → abandon, flag, return at minute 75.

FORMULA CARD — memorise cold in Block IV

Ch1: $(x+1/x)^2 - 2$ · $(x+1/x)^3 - 3(x+1/x)$ · $(a+b)^2 - 2ab$ · $(a+b)^2 - (a-b)^2 = 4ab$ · complete square: halve, square, subtract · digits = $\lfloor \log \rfloor + 1$

Ch2: $D = b^2 - 4ac$ (touch $\iff 0$, none $\iff <0$, distinct $\iff >0$) · sum = $-b/a$ · product = c/a · vertex = $-b/2a$ · both roots + $\iff$ sum>0, product>0, $D \geq 0$

Ch3: flip ONLY on ×÷ by negative · $|x| \leq a \iff -a \leq x \leq a$ · quadratic < 0 between roots · rational: sign chart + exclude denominator · "for all x" $\iff D<0$ & $a>0$

Ch4: + logs → multiply, − → divide · $\log_a a = 1$ · $a^b = N \iff b = \log_a N$ · chain collapse · digits formula · **domain first, reject negatives/zero**

Ch5: gcd × lcm = a·b · Euclid: divide until remainder 0 · same remainder $\Longrightarrow$ gcd of differences · $\tau = \Pi(e_i+1)$ · "or" count = A + B − both · pair form = gcd·x, gcd·y (coprime)

Ch6: reduce base mod n → reduce exponent mod cycle · last-digit cycles length 4 · heroes: $7^4 \equiv 01$, $3^{20} \equiv 01$ · inverse = multiply to $\equiv 1$ · $ax \equiv b$: divide by gcd first · consecutive-products proofs

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This is the whole plan. Tiers 1-2-3 in order, then playbook, then formula card. Nothing else exists tonight.